Indian Institute of Technology Patna
MA-538: M.Sc. II year
Spring Semester: 2023-24
Mid Semester Examination

Maximum Marks: 30 Total Time: 2 Hours

Note: Answer all six questions.

1. Suppose that $X_1, X_2, \ldots, X_n$ are independent and identically distributed (iid) random variables with common density function as $f_X(x) = (1/3), \ -2 \le x \le 1; \ = 0$, otherwise. Find density function of Y_n where $Y_n = \max(X_1, X_2, \ldots, X_n)$. Evaluate the conditional probability $P(-1 < Y_n \le (1/2) \mid -1.5 < Y_n \le 0)$. Also compute expectation of Y_n^2. **[2+1.5+1.5]**

2. Suppose that $X_1, X_2, \ldots, X_n$ are iid Poisson $P(\lambda)$ random variables. Show that it belongs to a one-parameter exponential family. Further show that $T = \sum_{i=1}^{n} X_i$ is a sufficient statistic for λ. Find also probability distribution of T using the MGF technique. Then find expectation of $e^{T/3}$. **[1+2+1.5+1.5]**

3. Suppose that X_1, X_2, X_3 and X_4 are iid uniform $U(0,1)$ distributed random variables. Compute the density function of the second order statistic Y_2. Then evaluate the probability $P((1/4) < Y_2 \le (3/4))$. Here Y_1, Y_2, Y_3 and Y_4 are corresponding order statistics. **[3 + 2]**

4. Suppose that X_1, X_2, X_3 and X_4 are iid random variables with common pdf $f_X(x) = e^{1-x}, \ 1 \le x < \infty; \ = 0$, otherwise. Now evaluate the probability $P(Y_1 > (5/4))$ from the density function of Y_1. Find the conditional expectation $E(e^{-Y_4} \mid Y_1 = 1)$. Here Y_1 and Y_4 are corresponding minimum and maximum order statistics respectively. **[2+4]**

5. Define bias of an estimator. Suppose that $X_1, X_2, \ldots, X_n$ are iid Bernoulli $Ber(p)$ random variables where p is unknown parameter. Use the C-R inequality to find UMVUE of p. **[1+3]**

6. Suppose that X_1 and X_2 are iid normal $N(\mu, 1)$ random variables. Find conditional distribution of X_1 given $X_1 + X_2 = t$ where t is any given constant. **[4]**